

SUPPLEMENTARY RESULTS: HIERTFR

This document contains the model-specific component and loss-function ablations for HierTFR, accompanying *Beyond the Majority Bin: A Methodological Framework for Balanced Automatic Pronunciation Assessment*. The cross-model comparisons and experimental reproducibility settings are reported in the paper’s supplementary material.

I. READING GUIDE

Baseline denotes the original model and training procedure. The +CV configuration introduces speaker-independent grouped cross-validation using `GroupKFold(n_splits=5)` without shuffling. Both +DB and +MS build on CV: +DB adds Dynamic Balanced MSE, whereas +MS adds Macro-MSE-based checkpoint selection. Complete BOMA combines CV, DB, and MS. In the loss-function ablations, CV and MS are held fixed while conventional MSE, DB-MSE, and Score-Balanced Loss (SB) are compared.

CCC-BS measures distribution-balanced agreement and PCC measures linear association; higher values are preferred for both. Micro-MSE and Macro-MSE are error measures on the unified $[0, 10]$ score scale, so lower values are preferred. Boldface marks the best value in each comparison. These metrics express complementary evaluation objectives; improvements under balanced criteria need not coincide with improvements under sample-weighted criteria.

II. RESULTS

For HierTFR, BOMA provides the highest CCC-BS on six aspects, while CV+MS is best for word total and utterance prosody and +DB is best for utterance completeness (Table S1). The Macro-MSE results favor BOMA on seven aspects, with one aspect each favoring +DB and +MS (Table S2). By contrast, the Baseline has the highest PCC on all nine aspects (Table S3). In the loss-only experiment, DB-MSE is best on seven CCC-BS aspects and eight Macro-MSE aspects (Tables S4 and S5).

TABLE S1
ABLATION STUDY FOR HIERTFR. CORRELATIONS ARE MEASURED USING CCC-BS.

Configuration	Phone Score	Word Score (CCC-BS)			Utterance Score (CCC-BS)				
	Phoneme $\uparrow$	Acc. $\uparrow$	Stress $\uparrow$	Total $\uparrow$	Acc. $\uparrow$	Comp. $\uparrow$	Fluency $\uparrow$	Prosody $\uparrow$	Total $\uparrow$
Baseline	0.425	0.560	0.105	0.596	0.737	0.010	0.683	0.685	0.778
+ CV	0.428	0.550	0.089	0.588	0.689	0.000	0.700	0.697	0.735
+ DB	0.542	0.658	0.566	0.548	0.721	0.064	0.750	0.729	0.764
+ MS	0.489	0.618	0.151	0.662	0.691	0.035	0.758	0.781	0.785
BOMA	0.613	0.665	0.625	0.612	0.768	0.048	0.781	0.768	0.800

^a The +DB and +MS configurations are evaluated on top of the CV configuration; BOMA combines CV, DB, and MS.

^b Bold values indicate the highest CCC-BS value for each aspect within this ablation.

TABLE S2
MICRO-MSE AND MACRO-MSE ABLATION STUDY FOR HIERTFR. LOWER VALUES INDICATE BETTER PERFORMANCE.

Configuration	Phone		Word Score						Utterance Score									
	Phoneme		Acc.		Stress		Total		Acc.		Comp.		Fluency		Prosody		Total	
	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓
Baseline	2.063	16.755	2.062	10.980	0.188	10.185	1.556	9.198	1.131	3.199	0.263	28.722	0.773	5.300	0.804	4.996	1.031	3.561
+ CV	2.143	16.120	2.118	10.830	0.189	10.328	1.608	9.029	1.175	2.852	0.266	29.118	0.820	4.921	0.833	4.730	1.070	4.067
+ DB	5.504	8.444	6.041	5.325	1.283	4.277	3.837	7.121	1.498	2.403	0.281	25.686	0.928	4.117	0.995	4.000	1.258	3.475
+ MS	2.522	13.372	2.509	8.470	0.190	9.298	1.935	6.881	1.359	2.592	0.255	27.259	0.974	3.828	0.940	3.303	1.244	3.241
BOMA	6.935	6.656	6.934	5.371	1.748	3.677	4.760	5.588	1.635	2.333	0.272	24.659	0.995	3.585	0.978	3.480	1.409	2.974

^a Micro-MSE (Mic.) and Macro-MSE (Mac.) are reported on the unified $[0, 10]$ scale; lower values ($\downarrow$) indicate better performance.

^b The +DB and +MS configurations are evaluated on top of CV; BOMA combines CV, DB, and MS.

^c Bold values indicate the lowest error for each metric within the model-specific ablation.

TABLE S3
PCC ABLATION STUDY FOR HIERTFR. HIGHER VALUES INDICATE BETTER PERFORMANCE.

Configuration	Phone Score	Word Score (PCC)			Utterance Score (PCC)				
	Phoneme ↑	Acc. ↑	Stress ↑	Total ↑	Acc. ↑	Comp. ↑	Fluency ↑	Prosody ↑	Total ↑
Baseline	0.644	0.622	0.325	0.634	0.735	0.513	0.801	0.795	0.764
+ CV	0.611	0.600	0.254	0.615	0.724	0.032	0.779	0.776	0.750
+ DB	0.501	0.486	0.214	0.516	0.697	0.310	0.754	0.757	0.729
+ MS	0.584	0.574	0.297	0.590	0.712	0.196	0.764	0.765	0.740
BOMA	0.496	0.474	0.200	0.493	0.697	0.209	0.760	0.763	0.732

^a The +DB and +MS configurations are evaluated on top of CV; BOMA combines CV, DB, and MS.

^b Bold values indicate the highest PCC value for each aspect within the model-specific ablation.

TABLE S4
LOSS-FUNCTION ABLATION IN TERMS OF CCC-BS FOR HIERTFR. ALL CONFIGURATIONS USE THE SAME CROSS-VALIDATION AND MACRO-MSE-BASED MODEL-SELECTION PROTOCOL; ONLY THE TRAINING LOSS IS CHANGED.

Loss	Phone Score	Word Score (CCC-BS)			Utterance Score (CCC-BS)				
	Phoneme ↑	Acc. ↑	Stress ↑	Total ↑	Acc. ↑	Comp. ↑	Fluency ↑	Prosody ↑	Total ↑
MSE	0.489	0.618	0.151	0.662	0.691	0.035	0.758	0.781	0.785
DB-MSE	0.613	0.665	0.625	0.612	0.768	0.048	0.781	0.768	0.800
SB	0.514	0.612	0.126	0.682	0.675	0.034	0.702	0.745	0.758

^a MSE corresponds to the CV+MS configuration, i.e., the BOMA validation and model-selection protocol trained with conventional MSE.

^b DB-MSE corresponds to the proposed Dynamic Balanced MSE used in the complete BOMA framework.

^c SB denotes Score-Balanced Loss, which replaces DB-MSE while keeping the remaining protocol unchanged.

^d Bold values indicate the highest CCC-BS for each aspect within the model-specific loss ablation.

TABLE S5
LOSS-FUNCTION ABLATION IN TERMS OF MICRO-MSE AND MACRO-MSE FOR HIERTFR. ALL CONFIGURATIONS USE THE SAME CROSS-VALIDATION AND MACRO-MSE-BASED MODEL-SELECTION PROTOCOL; ONLY THE TRAINING LOSS IS CHANGED.

	Phone		Word Score						Utterance Score								
	Phoneme		Acc.		Stress		Total		Acc.		Comp.		Fluency		Prosody		Total
Loss	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mac. ↓
MSE	2.522	13.372	2.509	8.470	0.190	9.298	1.935	6.881	1.359	2.592	0.255	27.259	0.974	3.828	0.940	3.303	1.244 3.241
DB-MSE	6.935	6.656	6.934	5.371	1.748	3.677	4.760	5.588	1.635	2.333	0.272	24.659	0.995	3.585	0.978	3.480	2.974
SB	2.686	12.360	2.645	8.106	0.186	9.722	2.012	6.337	1.381	2.576	0.251	27.251	0.957	4.670	0.922	3.900	1.294 3.505

^a MSE corresponds to the CV+MS configuration, i.e., the BOMA validation and model-selection protocol trained with conventional MSE.

^b DB-MSE corresponds to the proposed Dynamic Balanced MSE used in the complete BOMA framework.

^c SB denotes Score-Balanced Loss, which replaces DB-MSE while keeping the remaining protocol unchanged.

^d Micro-MSE (Mic.) and Macro-MSE (Mac.) are reported on the unified [0, 10] scale; lower values (↓) indicate better performance.

^e Bold values indicate the lowest error for each metric within the model-specific loss ablation.